Clean Hours Are Hot: The Cooling Penalty of Carbon-Aware Computing

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Abstract. Carbon-aware schedulers move flexible work to hours when the grid is cleanest. A data centre, however, draws IT power times its PUE, and PUE rises in hot weather. On solar-heavy grids the cleanest hours are also the hottest, so a scheduler that watches only carbon over-claims. With hourly weather at data-centre hubs in 47 cloud regions, carbon-only scheduling over-states its saving by 2.2% (efficient plants) to 6.9% (legacy plants) pooled, and by up to 61% in single regions. A closed form, $-\rho \cdot (\text{mean } c/\sigma_c) \cdot (\sigma_{PUE}/\text{mean PUE})$, predicts the penalty ($R^2$ 0.77-0.98) and, evaluated on the previous year, flags 62 of 67 cases above 10%. The same physics governs chilled-water storage: dispatched on grid carbon, as it is marketed, a tank keeps 1-26% of its value and raises emissions in up to 16 regions; dispatched on carbon × 1/COP it stacks fully on top of workload shifting.

1 Introduction

Temporal shifting of flexible compute is the most studied lever of carbon-aware computing [1, 2, 3]. Its benefit is computed from the grid's hourly carbon intensity c(t), and reported savings are usually scaled by an annual PUE. That treats the facility overhead as a constant. It is not: chillers reject heat to outdoor air or water, and their efficiency falls as the outdoor temperature rises. The carbon cost of one IT kWh is c(t)·PUE(t).

The two factors are correlated, and in the wrong direction. Solar makes midday both clean and hot. Figure 1 shows Johannesburg: the grid is cleanest around 11:00 UTC (13:00 local), exactly when a legacy plant's PUE peaks. Cooling-aware workload placement was studied a decade ago for single sites with on-site solar [4], and thermal storage for cost and peak shaving [5]; here we measure what ignoring cooling costs *grid*-carbon-aware schedulers across real grids and weather, and give a closed form for it.

Contributions.

- An exact accounting identity for the true saving, and the cooling penalty it implies (§3).
- Measurement on 47 cloud regions with ERA5 weather, 2020-2022, for two cooling plants (§4).
- A closed-form law that predicts the penalty out of sample from public data (§5).
- Chilled-water storage as a third battery: what drives it, and how it stacks with shifting (§6).

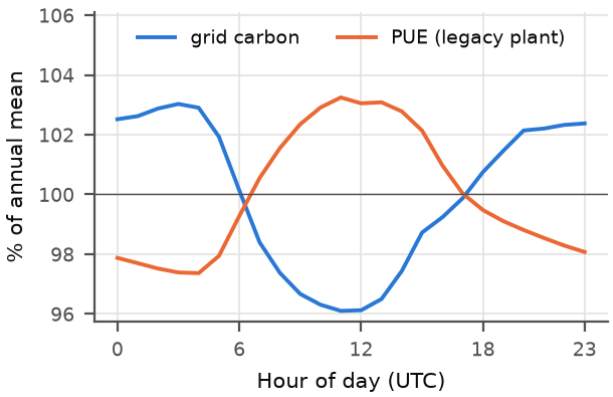


Figure 1. Mean day of grid carbon and PUE at Johannesburg (2022), each as a percent of its annual mean. The clean dip sits under the PUE peak.

2 Model

Each region gets one data-centre hub (e.g. Ashburn for PJM, Santa Clara for CAISO) with hourly 2-m dry- and wet-bulb temperature from ERA5 [7] and hourly carbon intensity from Electricity Maps [8], both in UTC. Two first-principles cooling plants bracket practice. *Hyperscale*: 18 °C chilled water, cooling tower on wet bulb, water-side economiser (free cooling below 10 °C wet bulb), 7% fixed overhead; median annual PUE 1.09. *Legacy*: 7 °C chilled water, air-cooled condenser on dry bulb, no economiser, 22% fixed overhead; median 1.40. Chiller COP is a fixed fraction of Carnot at the condensing temperature. The facility follows [9]: flat IT load, a share ϕ flexible within 24 h, IT power capped at 1 + h, solved as a year-long LP.

3 The Cooling Penalty

With the IT draw d scaled to mean 1, the saving against running on arrival is exactly **$-$cov(c·PUE, d)**. An operator who computes the saving on c and multiplies by the annual PUE claims $-$mean(PUE)·cov(c, d). The gap, **$-$cov(c·(PUE $-$ mean PUE), d)**, is the cooling penalty: positive when the schedule moves load into hours of above-average PUE. A scheduler can avoid it by optimising on c·PUE instead of c.

4 Measurement

In 72% of regions the mean daily profiles of carbon and PUE are anti-correlated ($\rho < -0.3$). With half the load flexible and +20% headroom, carbon-only scheduling over-claims by 2.2% (hyperscale) and 6.9% (legacy) pooled; the median region by 1.7% and 5.0%. The tail matters: 13 regions exceed 10% for legacy plants (Singapore 61%, Johannesburg 45%, Mumbai 42%). In Johannesburg the claim is 7.9 g/kWh and the truth 4.4.

Scheduling on c·PUE recovers 1.4% and 3.5% more true saving pooled. On grids whose carbon signal is flat (Hong

Kong, Jakarta), where carbon-only scheduling saves nothing, weather alone still saves up to 3.6 g/kWh.

Table 1. Cooling penalty as a share of the claimed saving, $\phi = 0.5$, 47 regions, each year's carbon and weather.

Pooled penalty	2020	2021	2022
Hyperscale, +20%	1.8%	2.2%	2.2%
Legacy, +20%	6.5%	7.4%	6.9%
Legacy, regions > 10%	15	13	13

5 A Closed-Form Law

If the schedule moves against the carbon signal, $\mathrm{cov}(\mathrm{PUE}, d)/\mathrm{cov}(c, d) \approx \rho \cdot \sigma_{\mathrm{PUE}}/\sigma_c$, so

$$\textbf{penalty / claim} \approx -\rho \cdot \textbf{(mean } c \textbf{ / } \sigma_c\textbf{)} \cdot \textbf{(}\sigma_{\mathbf{PUE}} \textbf{ / mean PUE)},$$

with ρ and the σs taken over within-day (25 h detrended) hourly variation. Daily-profile statistics are not enough (R^2 0.55-0.75); hourly residuals give R^2 0.77-0.98 against the LP over 12 configurations (Figure 2). The striking factor is mean c/σ_c, the inverse coefficient of variation of carbon: the most exposed grids are dirty ones with a small relative swing, where the cooling swing is large compared with the carbon swing. The law needs no optimiser. Evaluated on last year's data it predicts this year's LP penalty with R^2 0.79-0.90, and flags 62 of the 67 region-years above 10% (it runs slightly high, so it also raises some false alarms). Per-region penalties are stable: Spearman 0.83-0.95 between years.

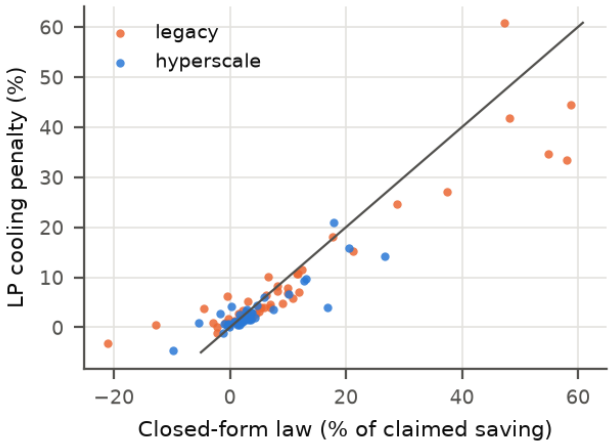


Figure 2. Cooling penalty from the LP against the closed-form law, $\phi = 0.5$, +20% headroom, one point per region and plant.

6 The Third Battery: Chilled-Water Storage

A chilled-water tank lets the plant make cold before it is needed. Cold costs $m(t) = 1/\mathrm{COP}$ units of electricity (zero under free cooling), so it should be made when $c \cdot m$ is low. We solve a cyclic LP per region: a tank of 1-12 h of heat load, chiller 1.5× peak, 1%/day standing loss. A 4 h tank cuts facility emissions by a median 0.23% (hyperscale) and 0.72% (legacy); cooling is only 13% of a legacy facility's emissions.

The tank is a weather battery. Dispatched on the weather alone it keeps 84% and 64% of its value; dispatched on grid carbon alone, as vendors propose, 1% and 26%, and it raises emissions in 16 and 9 of 47 regions (Figure 3), because it makes cold at clean, hot middays. One kWh of tank saves as much as 0.08-0.31 kWh of 4 h Li-ion battery, but on grids with a small daily spread, where a battery's losses eat its gain, the tank still saves. On top of cooling-aware shifting it adds its full saving (pooled ratio 1.00).

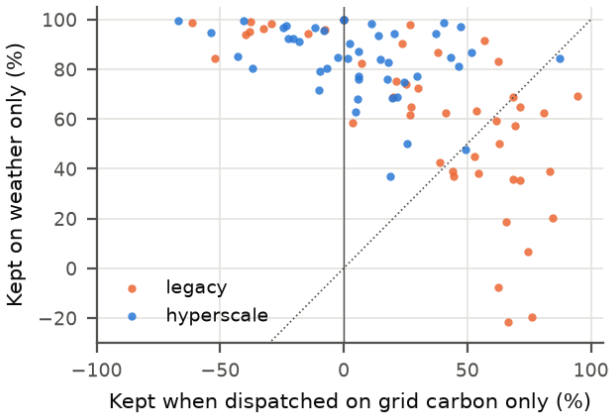


Figure 3. Share of a 4 h tank's saving kept when it is dispatched on grid carbon only or on weather only, one point per region and plant.

7 Discussion and Limitations

Implications. Carbon-aware schedulers should optimise on $c \cdot \mathrm{PUE}$, which every operator already meters, and report savings with hourly, not annual, PUE. Regions where the law exceeds a few percent can be identified from public carbon and weather data before deployment. The penalty concentrates in hot, humid, fossil-heavy regions (India, Southeast Asia, South Africa). Thermal storage should be dispatched on carbon × 1/COP; marketing it as 'charge on clean power' is wrong in a third of regions for efficient plants.

Limitations. Steady-state hourly plant models with assumed parameters (two archetypes bracket practice); one hub per region; PUE independent of IT load; average, not marginal, carbon; perfect forecasts; tank embodied carbon not counted.

References

[1] T. Sukprasert, A. Souza, N. Bashir, D. Irwin, P. Shenoy. On the Limitations of Carbon-Aware Temporal and Spatial Workload Shifting in the Cloud. EuroSys, 2024.

[2] A. Radovanovic, R. Koningstein, et al. Carbon-Aware Computing for Datacenters. arXiv:2106.11750, 2021.

[3] P. Wiesner, I. Behnke, D. Scheinert, K. Gontarska, L. Thamsen. Let's Wait Awhile: How Temporal Workload Shifting Can Reduce Carbon Emissions in the Cloud. Middleware, 2021.

[4] Z. Liu, Y. Chen, C. Bash, A. Wierman, D. Gmach, Z. Wang, M. Marwah, C. Hyser. Renewable and Cooling Aware Workload Management for Sustainable Data Centers. SIGMETRICS, 2012.

[5] W. Zheng, K. Ma, X. Wang. Exploiting Thermal Energy Storage to Reduce Data Center Capital and Operating Expenses. HPCA, 2014.

[6] A. Souza, N. Bashir, J. Murillo, W. Hanafy, Q. Liang, D. Irwin, P. Shenoy. Ecovisor: A Virtual Energy System for Carbon-Efficient Applications. ASPLOS, 2023.

[7] H. Hersbach et al. The ERA5 Global Reanalysis. Q. J. R. Meteorol. Soc. 146, 2020 (accessed via the Open-Meteo archive API).

[8] Electricity Maps. Hourly carbon-intensity data, 2020-2022, as released with [1].

[9] N. K. Lakkaram. Your Data Centre Is Already a Battery: A Closed-Form Theory of Carbon-Aware Shifting under Power Caps. Workshop draft, 2026.